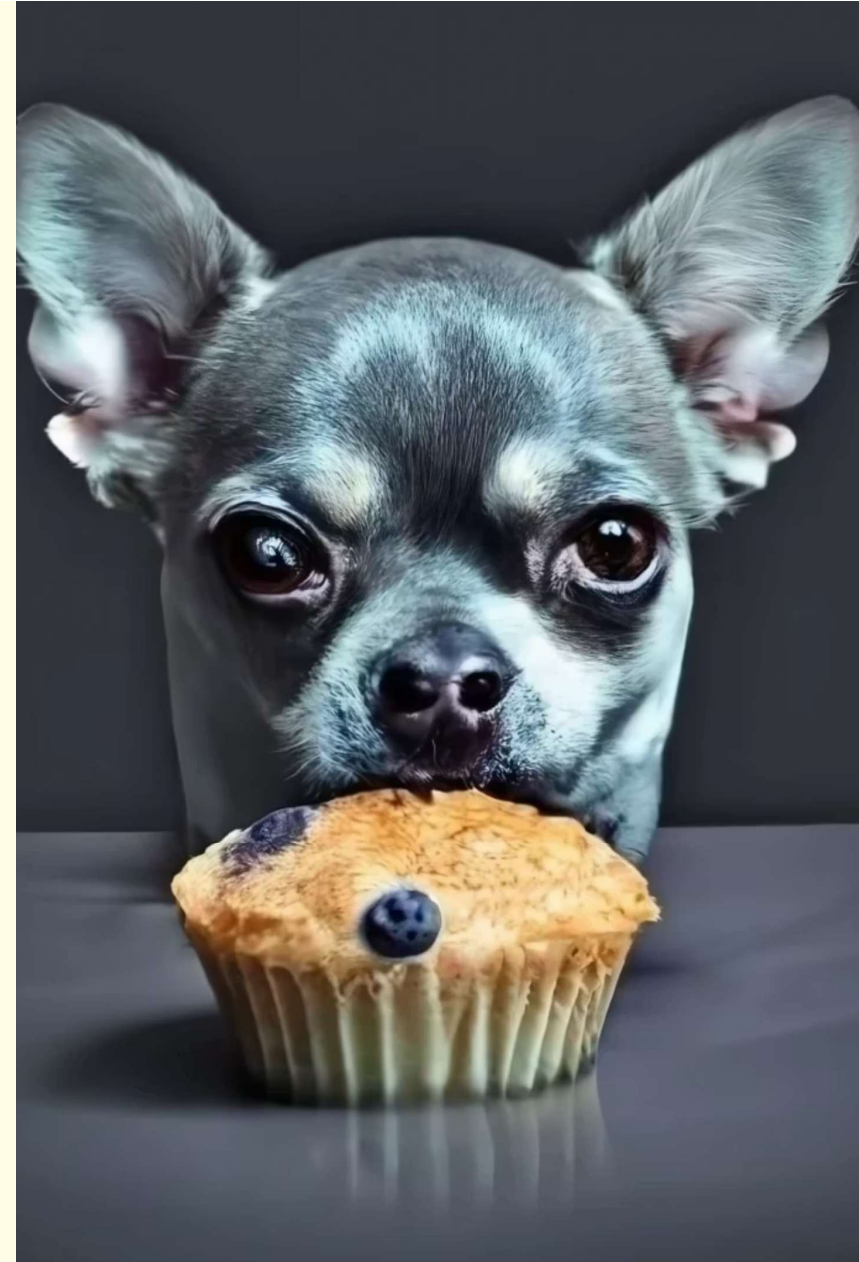


Data-Centric AI: Chihuahua vs Muffin Classification

TECH TITANS | Saveetha Engineering College



Problem & Objective



Classification Task

Binary classification: Chihuahua (0)
vs Muffin (1)



Primary Goal

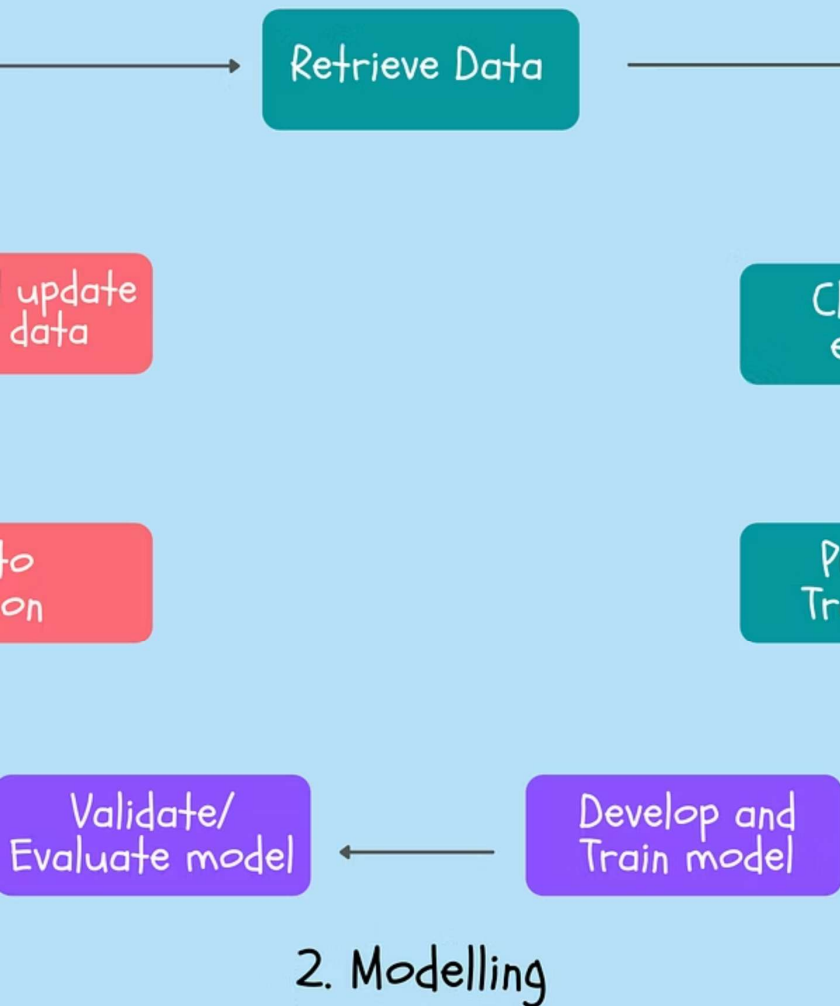
Improve model accuracy using data-
centric AI principles



Fixed Model

ResNet-18 architecture with 3LC
platform for labeling

Machine Learning Workflow



Dataset & Workflow

Dataset Composition

- 100 labeled training images
- 3,579 unlabeled images
- 1,000 validation images
- 1,184 hidden test images

Processing Pipeline

1. Train initial ResNet-18 model
2. Analyze UMAP embeddings
3. Label undefined samples
4. Retrain model
5. Generate predictions

3LC Data-Centric Approach

Embedding Visualization

UMAP embeddings for clustering analysis and pattern recognition

Active Learning

Strategic sample selection to maximize information gain

Sample Weighting

Confidence-based labeling to prioritize uncertain predictions

Powered by 3LC Dashboard for streamlined data management and model iteration

Results & Improvements



Validation Accuracy

Improved performance after relabeling iterations



Better Clustering

Enhanced embedding separation between classes



Confident Predictions

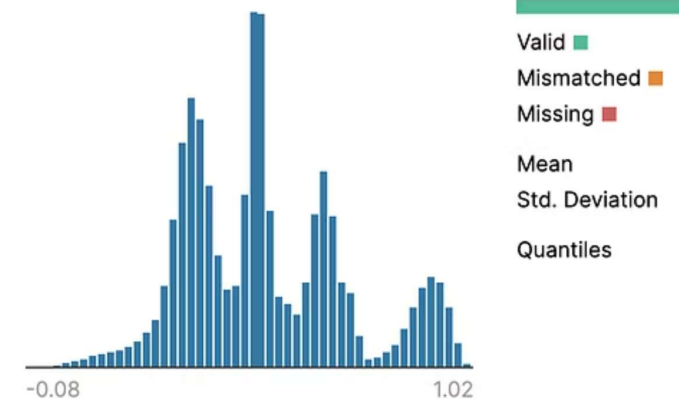
More reliable model outputs on test set

Submission.csv generated and uploaded to Kaggle leaderboard

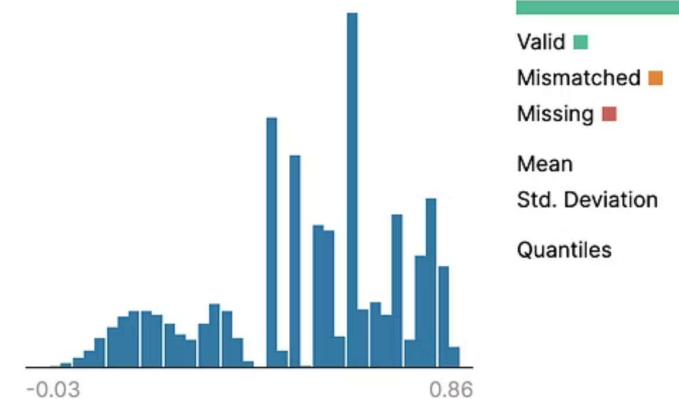
Discussion Leaderboard Datasets ...

My Submissions

cont1



cont2



Key Takeaways

Data Quality Matters

High-quality labels and curated datasets outperform complex model architectures

Embeddings Reveal Insights

Visualizing embeddings identifies mislabeled samples and ambiguous cases

Active Learning Efficiency

Strategic labeling improves performance with minimal annotation effort

Future Scope

- Additional labeling iterations
- Enhanced data balancing techniques
- Advanced data curation strategies

Thank You